

COLLEGE CODERS

React JS Lesson 9 – Lists and Keys (Premium Notes)

1) What are Lists in React

- Lists are used to display multiple items dynamically.
- Usually arrays are rendered into UI.

Example:

```
const fruits = ["Apple","Mango","Orange"];
```

Explanation:

- Array stores multiple values.

2) Rendering UI using .map()

- map() converts array data into JSX elements.

Example:

```
{fruits.map((item)=>(  
  <h2>{item}</h2>  
))}
```

Explanation:

- Each array item becomes one UI element.

3) What are Keys & Why They Are Needed

- Keys help React identify each item uniquely.
- React uses keys during updates.

Example:

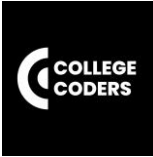
```
{fruits.map((item,index)=>(  
  <h2 key={index}>{item}</h2>  
))}
```

Explanation:

- Without key React gives warning.

4) Index as Key vs Unique Key

- Index works but unique id is better.



COLLEGE CODERS

Example:

```
key={item.id}
```

Explanation:

- Unique key prevents wrong updates when list changes.

5) Adding Items to a List

- Use spread operator to add new item.

Example:

```
setFruits([...fruits,"Banana"]);
```

Explanation:

- Old array copied + new item added.

6) Removing First Item (splice method)

- Remove first element carefully using copied array.

Example:

```
let arr=[...fruits];  
arr.splice(0,1);  
setFruits(arr);
```

Explanation:

- Original state should not mutate directly.

7) Removing Specific Item (splice with index)

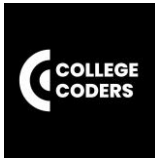
- Use selected index.

Example:

```
let arr=[...fruits];  
arr.splice(index,1);  
setFruits(arr);
```

Explanation:

- Selected item removed.



COLLEGE CODERS

8) Updating Array State Correctly

- Always create new array before updating.

Example:

```
setFruits(fruits.map(item=>item));
```

Explanation:

- React needs new reference.

9) Passing Parameters in Event Handlers

- Arrow function sends custom index/value.

Example:

```
<button onClick={()=>removeItem(index)}>Delete</button>
```

Explanation:

- Index reaches remove function.

Mini Project – Dynamic Fruit List

- Add fruit using button.
- Remove first fruit.
- Delete selected fruit.